

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Computer Applications (B.C.A.)

Semester-III University Examination March-2018

CA 214 / CA 203 Object Oriented Programming using C++

Date: 28.03.2018, Wednesday Time: 01.30 pm to 04.30 pm Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION-I

Q.1 Attempt the following.

[11]

1. A single line comment begin with // in a C++ source code. True / False
2. C++ was developed by _____.
(a) Bjarne Stroustrup (b) Dennis Ritchie (c) James A. Gosling (d) E. F. Codd
3. The member function can be defined outside the class using _____ operator.
(a) ?: ternary operator (b) : colon (c) :: scope resolution (d) none of these
4. Destructor receives zero argument. True / False
5. _____ means to determine at run time which method to be invoked.
(a) Dynamic hiding (b) Dynamic typing (c) Dynamic loading (d) Dynamic binding
6. _____ keyword is used to overload operators.
(a) operators (b) overload (c) operator (d) operand
7. Reference variable provides an alternative name for a previously defined variable.
True / False
8. In C++, _____ header file includes input/output operations.
(a) conio (b) stdlib (c) math (d) iostream
9. By default, inheritance is always private. True / False
10. _____ operator is used with cout.
(a) >> (b) > (c) << (d) <
11. The full form of MIL is _____.
(a) Member Initialization List (b) Member Inspection List
(c) Media Integration Layer (d) Microsoft Intermediate Language

Q.2 [A] Explain any six concepts of object oriented programming in detail.

[06]

[B] Differentiate between constructor and destructor.

[06]

OR

Q.2 [A] What is called operator overloading? List out all the rules of operator overloading.

[06]

[B] Differentiate between POP and OOP.

[06]

Q.3 Answer following questions in detail (Any three)

[12]

1. Write shortnote on function overloading.
2. Write a C++ program to overload binary + and – operators for Integer class.
3. What is access specifier? Explain in detail with example.
4. What is friend function? Explain with suitable example.
5. What is constructor? List down characteristics of constructor.

[P.T.O]

SECTION-II

Q.4 Attempt the following.

[11]

1. ifstream, ofstream and fstream classes are declared in a fstream header file. True / False
2. _____ means wrapping up of data and functions into a single unit.
(a) Data encapsulation (b) Polymorphism (c) Inheritance (d) Abstraction
3. Object can not be created from abstract class. True / False
4. The expression used to catch all is _____.
(a) catch (b) catch(...) (c) try (d) try(...)
5. _____ means deriving a new class from the existing class.
(a) Abstraction (b) Polymorphism (c) Data encapsulation (d) Inheritance
6. Abstraction provides concept of reusability. True / False
7. _____ are peculiar problems that a program may encounter at run time.
(a) Errors (b) Bugs (c) Exceptions (d) Garbage
8. A base class is made _____ to avoid duplication in inheritance.
(a) virtual (b) abstract (c) super (d) none of the above
9. The try block is executed when a specific exception is thrown. True / False
10. Which of the following is not a valid access specifier?
(a) public (b) private (c) protected (d) mutable
11. _____ operator can never be overloaded.
(a) / (b) () (c) :: (d) %

Q.5 [A] Define Inheritance. Explain various forms of inheritance.

[06]

[B] Which are the ways in which a file can be opened? Give one example for each.

[06]

OR

Q.5 [A] What is virtual base class? Explain why it is needed? Give one example.

[06]

[B] How is an exception handled in c++? Explain with suitable example.

[06]

Q.6 Answer following questions in detail (Any three)

[12]

1. Explain multiple inheritance with example.
 2. Explain main three components of Exception handling.
 3. Write a C++ program to implement multilevel inheritance.
 4. When do we use multiple catch handlers? Explain with example.
 5. Explain seekg(), seekp(), tellg() and tellp() functions for manipulation of file pointers.
-